

Kinematics

Question 1 (Conceptual): Two cars, A and B, start from rest and accelerate uniformly. Car A accelerates at 2 m/s^2 for 10 s, while Car B accelerates at 4 m/s^2 for 5 s.

Which car travels farther by the end of its acceleration phase? Explain your reasoning in terms of the kinematic equation $s = v_0 t + \frac{1}{2} a t^2$.

Question 2 (Calculation): A cyclist starts from rest and accelerates uniformly to a speed of 20 m/s over a distance of 200 m.

- (a) Identify the appropriate kinematic equation.
- (b) Calculate the cyclist's acceleration.
- (c) Determine the time taken to reach 20 m/s.